Prosthetic Control Interface Board

PCB Design Internship Assignment Report

Tufan Layek

B.Tech Electronics and Communication Engineering
SRM Institute of Science and Technology

Tool Used: EasyEDA
Board Type: 2-Layer PCB
Date: May 21, 2026

Contents

1 Project Overview

1.1 Introduction

This project presents the design and development of a compact 2-layer embedded control PCB named **Prosthetic Control Interface Board**. The board is designed to interface biological and mechanical sensing systems with an embedded microcontroller for real-time signal processing and actuator control applications.

The system integrates multiple functional hardware blocks including power regulation, sensor interfacing, embedded processing, actuator driving, communication headers, and USB programming support into a single manufacturable PCB. The design focuses on clean schematic organization, efficient PCB layout, proper routing practices, and prototype-ready hardware implementation.

The PCB is built around the ESP32-WROOM-32 microcontroller module, which provides processing capability along with wireless communication support. The board accepts sensor inputs from an EMG sensor and a flex sensor, processes the acquired signals, and controls external actuators using MOSFET-based driver circuits.

The project was developed using EasyEDA and follows standard PCB design practices including ground plane implementation, differential signal routing, proper trace width management, decoupling techniques, and component placement optimization.

1.2 Objective

The primary objective of the system is to acquire human motion and muscle activity signals using EMG and flex sensors, process these signals through the ESP32 microcontroller, and generate control signals for external actuators or motors used in prosthetic movement applications.

The EMG sensor is used to detect muscle activation signals from the user, while the flex sensor is used to measure finger or joint bending movement. By combining both sensor inputs, the system can achieve improved motion interpretation and more reliable prosthetic control behavior.

The processed sensor data can be used to drive motors or actuators responsible for prosthetic finger movement, gripping action, or assistive motion control.

This project is to design a functional and manufacturable embedded control PCB capable of interfacing sensors and controlling external actuators in prosthetic or embedded automation applications.

The project aims to demonstrate practical understanding of:

- Electronic schematic design
- PCB layout and routing
- Embedded hardware integration
- Power distribution and regulation
- Sensor interfacing techniques
- MOSFET-based output control
- USB communication and programming interface design
- Ground plane and signal integrity considerations

The design also focuses on creating a compact prototype-ready PCB suitable for hardware development, testing, and future system expansion.

1.3 System Overview

The Prosthetic Control Interface Board consists of multiple interconnected hardware blocks that collectively perform sensing, processing, communication, and actuator control functions.

- **EMG Sensor Interfacing:** The board includes an analog input interface for an EMG sensor module used to acquire muscle activity signals. The signal is routed to the ESP32 ADC input for processing.
- **Flex Sensor Interfacing:** A dedicated flex sensor input section is provided to detect bending or movement-based input signals. Proper pull-down resistor configuration is used to stabilize the sensor output.
- **ESP32 Processing Unit:** The ESP32-WROOM-32 microcontroller acts as the central processing unit of the system. It handles sensor data acquisition, signal processing, actuator control, and communication tasks.
- **Motor / Actuator Control:** The PCB includes MOSFET-based output driver circuits capable of controlling external loads such as DC motors, actuators, or relays. Flyback diodes are included for inductive load protection.
- **USB Programming Interface:** A USB Type-C interface along with the CH340C USB-to-UART converter is implemented to enable firmware uploading and serial communication with the ESP32.

2 System Architecture

The overall system architecture of the Prosthetic Control Interface Board is designed around the ESP32-WROOM-32 microcontroller, which acts as the central processing and control unit of the system. The architecture integrates sensor acquisition, embedded signal processing, actuator control, communication interfaces, and power management into a compact and modular PCB platform.

The EMG sensor and flex sensor provide analog input signals corresponding to muscle activity and finger bending motion. These signals are processed by the ESP32 microcontroller to generate appropriate control outputs for external actuators or motors through MOSFET-based driver circuits. The board also includes communication headers for UART and I^2C interfacing, enabling debugging and future peripheral expansion.

Power regulation and protection circuitry ensure stable 3.3V operation for the embedded system, while the USB Type-C programming interface with CH340C provides firmware uploading and serial communication capability.

The complete functional relationship between different hardware modules is illustrated in Figure 1.

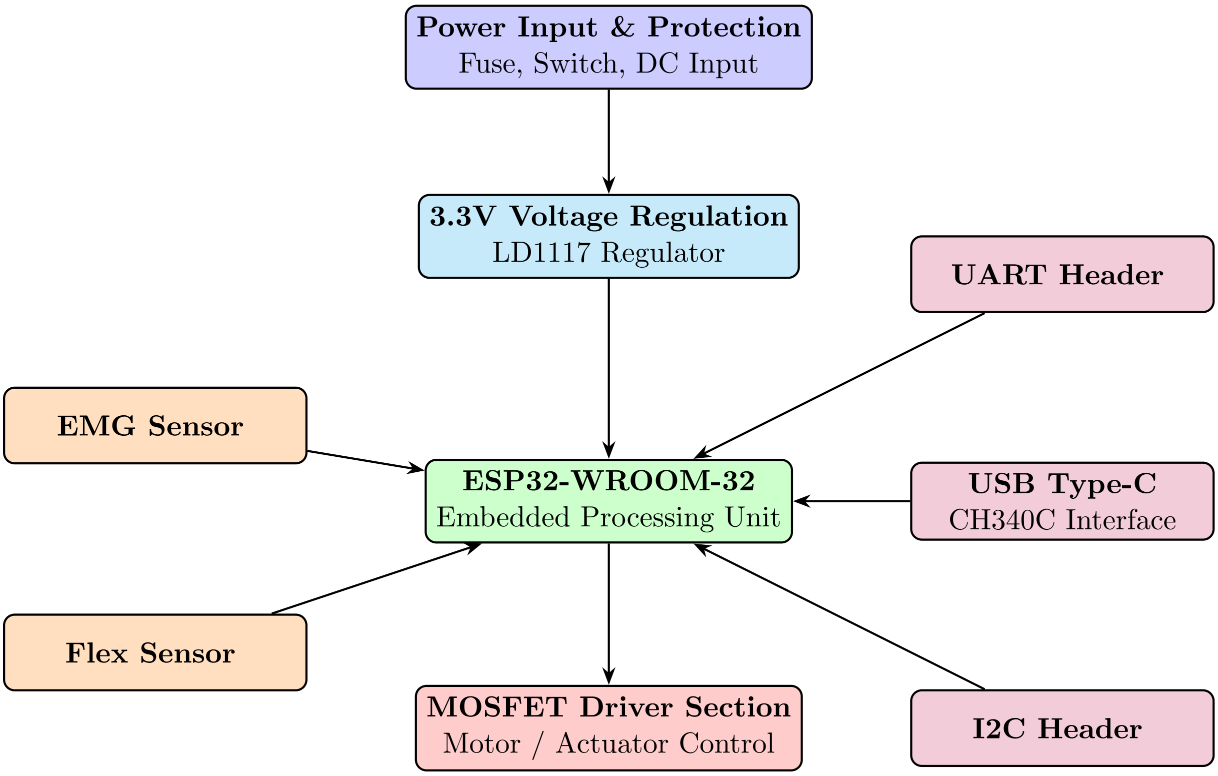

Figure 1: Functional Block Diagram of Prosthetic Control Interface Board

3 Circuit Design

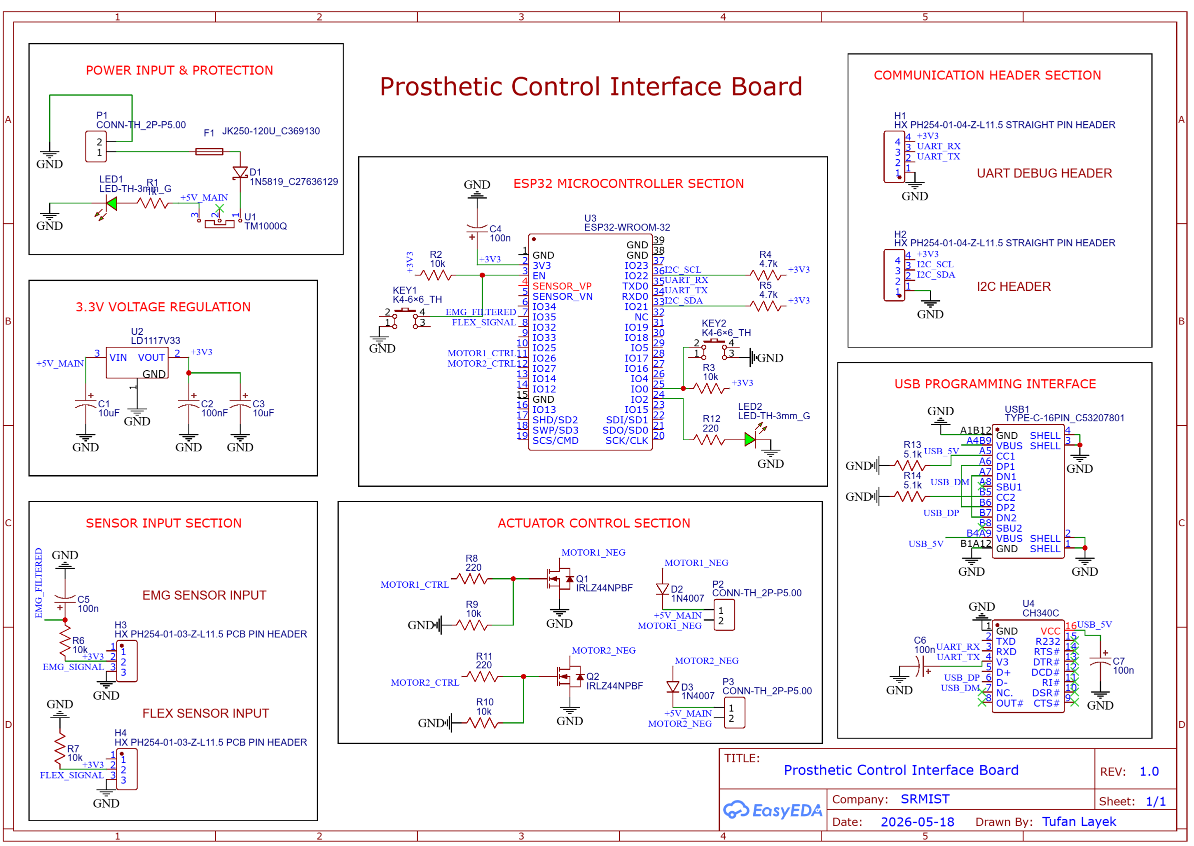

Figure 2: Complete Schematic of the Prosthetic Control Interface Board

3.1 Power Input and Protection Section

The Power Input and Protection Section is responsible for supplying stable input power to the PCB while protecting the circuit from electrical faults and incorrect power connections. A DC barrel connector is used as the primary input source for the board. The incoming supply first passes through a fuse for overcurrent protection. In the event of excessive current flow or short circuit conditions, the fuse disconnects the circuit and prevents component damage.

A Schottky diode is connected in series with the input supply to provide reverse polarity protection. This prevents current flow if the external power supply is connected with incorrect polarity. A power switch is included to manually control the power supplied to the entire board.

An LED indicator along with a current limiting resistor is used as a power status indicator. The LED glows whenever the board receives power, allowing easy visual confirmation of power availability illustrated in Figure 2

3.2 Voltage Regulation Section

The Voltage Regulation Section generates a stable 3.3V supply required by the ESP32 microcontroller and associated low-voltage circuitry. The regulation stage is implemented using the LD1117-3.3 linear voltage regulator.

The regulator converts the higher DC input voltage into a regulated 3.3V output suitable for embedded electronics. An electrolytic capacitor is connected at the input side of the regulator to reduce voltage fluctuations and supply ripple. Additional output capacitors are connected at the regulator output to improve voltage stability and suppress high-frequency noise.

The regulated 3.3V supply is distributed throughout the PCB using dedicated power traces and is used by the ESP32 module, sensor interface circuitry, and communication headers illustrated in Figure 2

3.3 ESP32 Microcontroller Section

The ESP32 Microcontroller Section acts as the central processing unit of the Prosthetic Control Interface Board. The design utilizes the ESP32-WROOM-32 module due to its integrated Wi-Fi capability, dual-core processing architecture, low power consumption, and extensive GPIO support.

A reset push button is connected to the EN pin of the ESP32 to allow manual resetting of the microcontroller during programming and debugging operations. Pull-up resistors are connected to important control lines to ensure stable logic levels during startup and operation.

A status LED is connected to one of the GPIO pins through a current limiting resistor. The LED can be controlled through firmware for status indication, debugging, and activity monitoring.

Several GPIO pins are allocated for sensor inputs, communication interfaces, and actuator control outputs. Dedicated ADC pins are used for analog sensor acquisition, while digital GPIO pins are used for MOSFET gate control and communication interfaces such as UART and I2C illustrated in Figure 2

3.4 Sensor Input Section

The Sensor Input Section provides interfaces for EMG and flex sensors used for motion and muscle activity detection. Both sensor inputs are connected to the analog input pins of the ESP32 microcontroller.

The EMG sensor interface receives analog muscle activity signals that are processed by the

ESP32 ADC. Similarly, the flex sensor interface detects bending variations and converts them into analog voltage signals for processing.

Pull-down resistors are connected to the sensor signal lines to prevent floating voltage conditions when the sensors are disconnected or inactive. Capacitors are also included for basic signal filtering and noise reduction.

The analog sensor signals are routed carefully to minimize interference from high-current switching sections of the PCB illustrated in Figure 2

3.5 Actuator Control Section

The Actuator Control Section is responsible for driving external loads such as motors or actuators. The circuit uses N-channel MOSFETs as electronic switches controlled by the ESP32 GPIO pins.

The MOSFET gates are connected through gate resistors to limit switching current and improve signal stability. Pull-down resistors are connected to the MOSFET gates to ensure that the transistors remain OFF during startup conditions.

Flyback diodes are connected across the output terminals to suppress voltage spikes generated by inductive loads during switching operations. This protects the MOSFETs and improves overall circuit reliability.

Dedicated output connectors are provided for connecting external motors or actuators to the board illustrated in Figure 2

3.6 Communication Header Section

The Communication Header Section provides external communication and debugging interfaces for system expansion and development purposes.

A UART header is included for serial communication, firmware debugging, and real-time data monitoring. The UART interface can be connected to external serial converters or debugging tools.

An I2C communication header is also provided for interfacing additional peripherals such as sensors, displays, EEPROMs, or communication modules. Pull-up resistors are connected to the SDA and SCL lines to ensure reliable I2C communication.

These communication headers improve system flexibility and allow future hardware expansion illustrated in Figure 2

3.7 USB Programming Interface

The USB Programming Interface enables firmware uploading and serial communication with the ESP32 microcontroller. A USB Type-C connector is used as the physical interface for modern connectivity and improved usability.

The CH340C USB-to-UART converter IC is used to convert USB communication into UART signals compatible with the ESP32 programming interface. The TX and RX lines of the CH340C are connected directly to the ESP32 UART pins.

The USB differential pair lines (D+ and D-) are routed carefully on the PCB to maintain signal integrity and reduce communication errors. Proper trace width, spacing, and routing practices are followed to ensure reliable USB communication.

This section allows easy programming, debugging, and serial monitoring of the embedded system through a standard USB connection illustrated in Figure 2

4 Design Calculations

4.1 Voltage Regulator Calculation

The LD1117-3.3 voltage regulator is used to generate a regulated 3.3V supply from the main DC input voltage.

$$V_{OUT} = 3.3V$$

Assuming:

$$V_{IN} = 5V$$

Voltage drop across the regulator:

$$V_{DROP} = V_{IN} - V_{OUT}$$

$$V_{DROP} = 5V - 3.3V = 1.7V$$

The regulator operates within the dropout voltage specification of the LD1117 regulator.

4.2 LED Current Limiting Resistor Calculation

The indicator LEDs use current limiting resistors to prevent excessive current flow.

Using Ohm's Law:

$$R = \frac{V_S - V_F}{I}$$

Where:

- $V_S = 3.3V$
- $V_F = 2.0V$ (Typical LED forward voltage)
- $I = 10mA = 0.01A$

Therefore,

$$R = \frac{3.3 - 2.0}{0.01}$$

$$R = 130\Omega$$

A standard resistor value of 220Ω was selected to reduce LED current and power consumption.

4.3 MOSFET Gate Pull-Down Resistor

A 10kΩ pull-down resistor is connected to the MOSFET gate to ensure the MOSFET remains OFF during ESP32 startup or floating GPIO conditions.

The pull-down current is:

$$I = \frac{V}{R}$$

$$I = \frac{3.3V}{10k\Omega}$$

$$I = 0.33mA$$

This current is sufficiently small while maintaining stable logic levels.

4.4 Capacitor Selection

Decoupling capacitors were added near the ESP32 and voltage regulator to suppress high-frequency noise and improve power stability.

- $100nF$ capacitors used for high-frequency decoupling
- $10uF$ capacitors used for bulk filtering and voltage stabilization

4.5 PCB Trace Width Calculations

PCB trace widths were selected based on the expected current carrying requirements of different sections of the circuit. Wider traces were used for high-current power and motor lines, while thinner traces were used for low-current signal routing.

The calculations were estimated using standard IPC-2221 [4] PCB trace current guidelines for external copper layers with 1 oz copper thickness.

The empirical IPC-2221 equation for external trace width estimation is:

$$I = k \times (\Delta T)^{0.44} \times A^{0.725}$$

Where:

- I = Current in Amperes
- $k = 0.048$ for external layers
- ΔT = Allowed temperature rise ($^\circ C$)
- A = Cross-sectional area of trace in mil^2

For 1 oz copper PCB:

$$\text{Copper Thickness} = 1.37 \text{ mil}$$

The trace width is obtained from:

$$W = \frac{A}{T}$$

Where:

- W = Trace width (mil)
- A = Cross-sectional area (mil^2)
- T = Copper thickness (mil)

4.5.1 Main Power Trace Width

The main power input traces carry current for the complete PCB including the ESP32, sensors, and actuator control section.

Assuming:

$$I = 1A$$

Allowed temperature rise:

$$\Delta T = 10°C$$

Using IPC-2221 approximation:

$$A \approx 64 \text{ mil}^2$$

Therefore,

$$W = \frac{64}{1.37}$$

$$W \approx 46.7 \text{ mil}$$

Converting to millimeters:

$$W \approx 1.18 \text{ mm}$$

Hence, a practical trace width of:

$$\boxed{1.2 \text{ mm}}$$

was selected for the main power traces.

4.5.2 Motor Output Trace Width

The MOSFET output traces driving external actuators require larger trace widths because they may carry higher switching currents.

Assuming:

$$I = 1.5A$$

Using IPC-2221 estimation:

$$A \approx 85 \text{ mil}^2$$

Therefore,

$$W = \frac{85}{1.37}$$

$$W \approx 62 \text{ mil}$$

Converting to millimeters:

$$W \approx 1.57 \text{ mm}$$

Thus, the selected motor trace width was:

$$\boxed{1.5 \text{ mm}}$$

4.5.3 3.3V Power Trace Width

The regulated 3.3V lines supply the ESP32 and low-power peripherals.

Assuming:

$$I = 0.3A$$

Using IPC approximation:

$$A \approx 30 \text{ mil}^2$$

Therefore,

$$W = \frac{30}{1.37}$$

$$W \approx 21.9 \text{ mil}$$

Converting to millimeters:

$$W \approx 0.56 \text{ mm}$$

Hence, a standard width of:

$$\boxed{0.6 \text{ mm}}$$

was selected for the 3.3V power traces.

4.5.4 Sensor Signal Trace Width

The EMG and flex sensor signals carry extremely low current analog signals.

For low-current analog signal routing, thin traces are sufficient while helping reduce routing congestion.

Therefore, a trace width of:

$$\boxed{0.25 \text{ mm}}$$

was selected for sensor signal lines.

4.5.5 USB Differential Pair Width

The USB D+ and D lines were routed as a differential pair.

For USB Full-Speed communication:

- Equal trace lengths were maintained
- Parallel routing was used
- Minimal vias were used
- Controlled spacing was maintained

A trace width of:

$$\boxed{0.2 \text{ mm}}$$

was selected for the USB differential pair routing.

4.5.6 Final Trace Width Summary

Net Type	Current Range	Selected Width
Main Power	Up to 1A	1.2 mm
Motor Output	Up to 1.5A	1.5 mm
3.3V Power	Up to 300mA	0.6 mm
Sensor Signals	Low Current	0.25 mm
USB Differential Pair	High-Speed Signal	0.2 mm

Table 1: PCB Trace Width Selection Summary

5 PCB Design and Layout

5.1 PCB Specifications

The Prosthetic Control Interface Board was designed as a compact and manufacturable 2-layer printed circuit board suitable for prototyping and embedded hardware development. The PCB layout was developed using EasyEDA while following standard PCB design guidelines for signal integrity, routing efficiency, and manufacturability.

The design utilizes a dual-layer structure consisting of a Top Layer and Bottom Layer for component placement and routing. A ground polygon pour was implemented on the bottom layer to improve grounding performance, reduce noise, and simplify return current paths.

The PCB was designed using standard FR4 substrate material with 1 oz copper thickness on both layers with compact component placement while maintaining adequate spacing between functional blocks. Proper routing techniques and trace width selection were used to support both low-current signal lines and higher-current actuator control traces. The final PCB dimensions are approximately $85\,\text{mm} \times 55\,\text{mm}$, making the design compact and suitable for prototype manufacturing and embedded hardware development.

Design Rule Check (DRC) and Electrical Rule Check (ERC) were performed in EasyEDA to verify schematic connectivity, routing clearances, trace spacing, via dimensions, and manufacturability constraints. All major routing violations and connectivity issues were resolved before finalizing the PCB layout.

The board was optimized to ensure compatibility with standard PCB fabrication processes and prototype assembly techniques.

5.2 Component Placement Strategy

Component placement was performed based on functional block organization and signal flow optimization. The PCB was divided into separate sections including power regulation, microcontroller processing, sensor interfacing, communication interfaces, and actuator driving circuitry.

The ESP32-WROOM-32 module was positioned centrally to reduce routing complexity between sensors, communication headers, and output driver circuits. Sensor interface components were placed near the analog input section of the ESP32 to minimize analog trace lengths and reduce noise pickup.

The actuator control section containing MOSFETs and output connectors was isolated from the sensor input section to minimize switching noise interference. Power regulation components were positioned close to the power input connector to reduce voltage drop and improve power stability.

Communication interfaces such as USB, UART, and I2C headers were placed near the PCB edges for easy accessibility during programming, debugging, and peripheral interfacing.

The overall placement strategy improved routing efficiency, signal integrity, and maintainability of the PCB design.

5.3 Routing Strategy

The routing strategy focused on achieving reliable electrical performance while maintaining a clean and manufacturable PCB layout. Different trace widths were selected depending on the current carrying requirements of individual nets.

Wider traces were used for the main power supply and actuator output paths to reduce resistive losses and improve current handling capability. Medium-width traces were used for regulated 3.3V power distribution, while thinner traces were used for low-current sensor and communication signals.

Special attention was given to USB differential pair routing. The USB D+ and D- traces were routed carefully with controlled spacing and approximately equal lengths to maintain signal integrity during USB communication.

A ground polygon pour was implemented on the bottom layer to provide a continuous low-impedance ground reference across the PCB. This improved return current flow, reduced electromagnetic interference, and simplified overall routing.

Vias were used wherever layer transitions were required during routing. Additional stitching vias were also added in selected locations to improve grounding continuity between PCB layers.

The routing strategy helped improve electrical reliability, reduce noise, and simplify PCB fabrication.

5.4 ESP32 Antenna Keepout

The ESP32-WROOM-32 module contains an onboard PCB antenna used for wireless communication. To ensure proper antenna performance, a keepout region was maintained below and around the antenna section of the module.

No copper pour, signal traces, vias, or components were placed directly beneath the antenna region. This minimizes RF signal absorption and prevents distortion of the antenna radiation pattern.

Proper antenna clearance improves wireless communication range, signal stability, and overall RF performance of the ESP32 module. Maintaining the keepout region is an important PCB design consideration for RF-enabled embedded systems.

5.5 PCB Layout Screenshots

The following figures 3,4,5,6 show the final PCB layout generated using EasyEDA. The layout demonstrates organized functional block placement, efficient routing, and ground plane implementation.

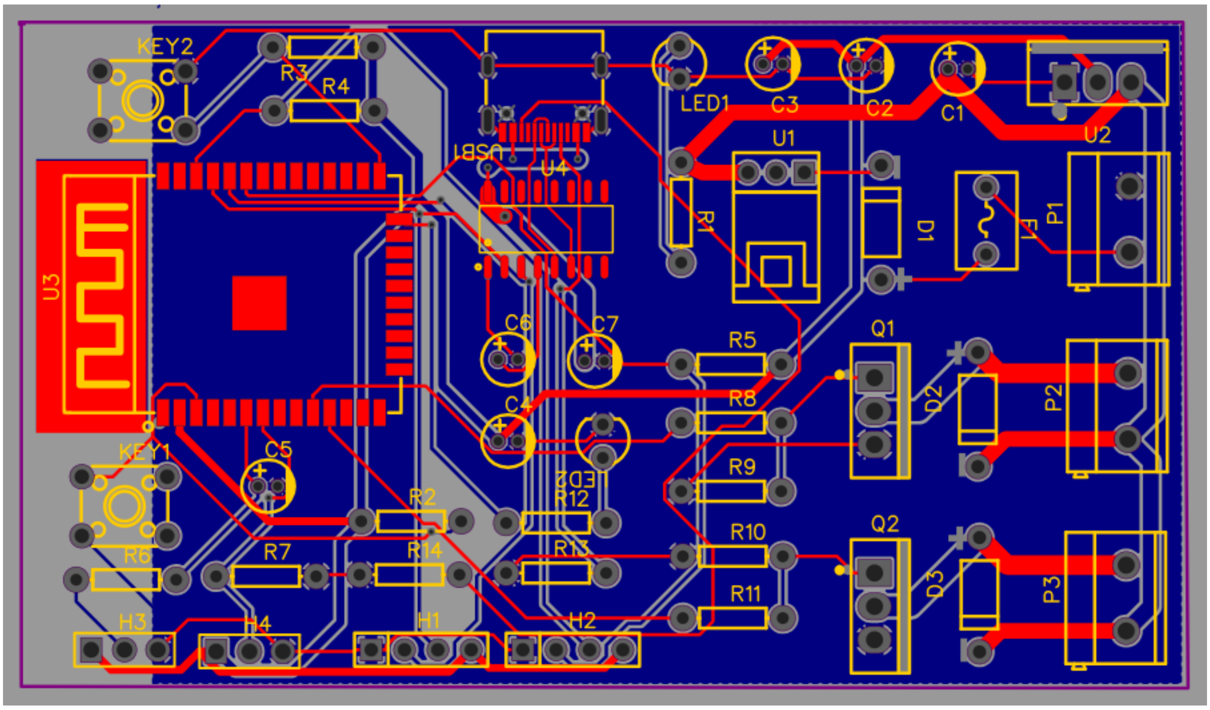

Figure 3: Top Layer PCB Layout

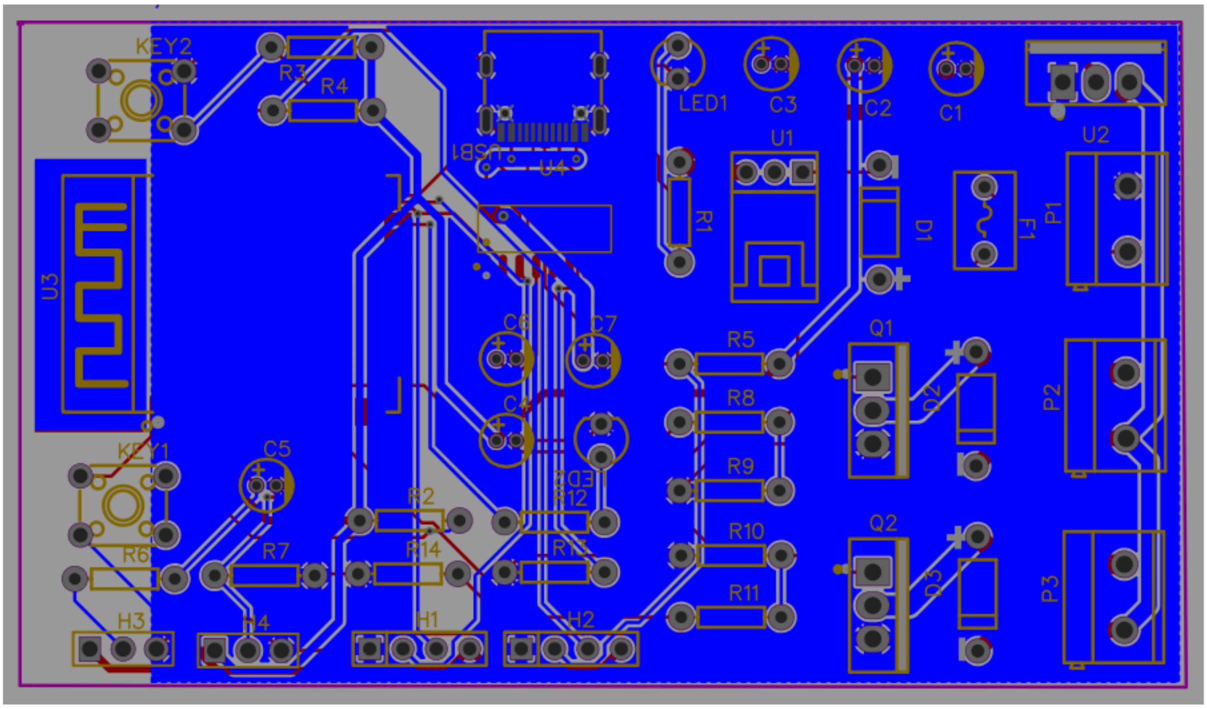

Figure 4: Bottom Layer PCB Layout

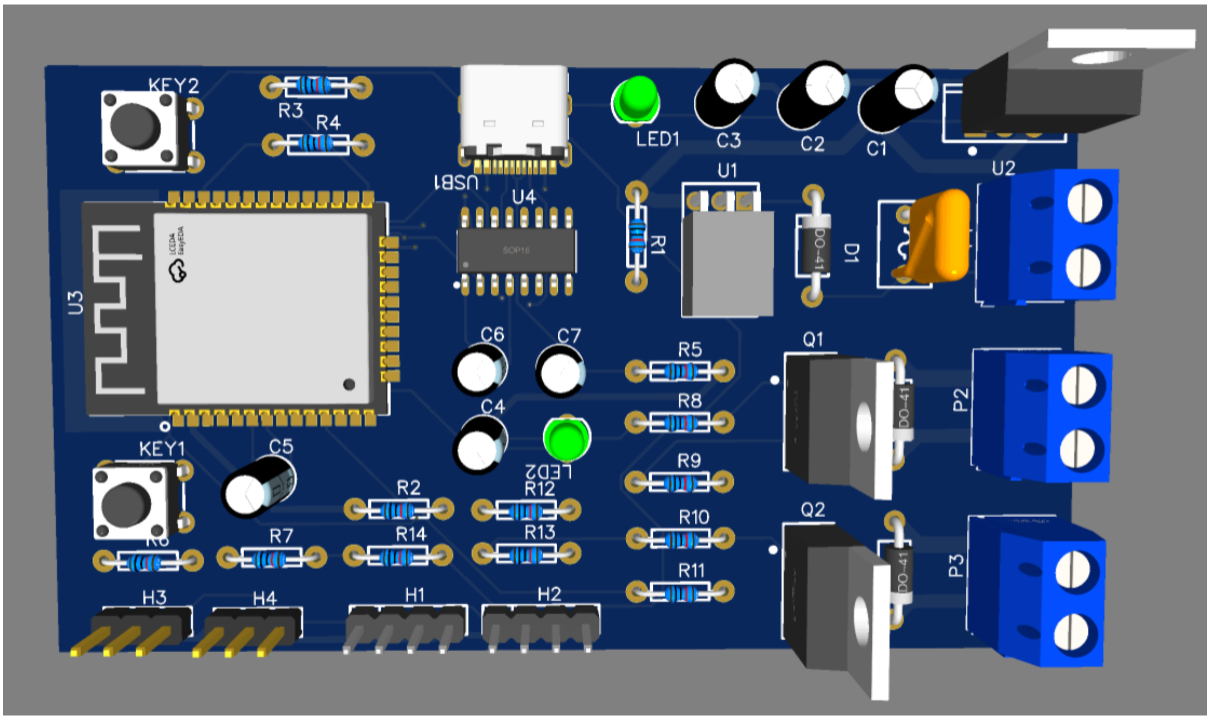

Figure 5: 3D Top View

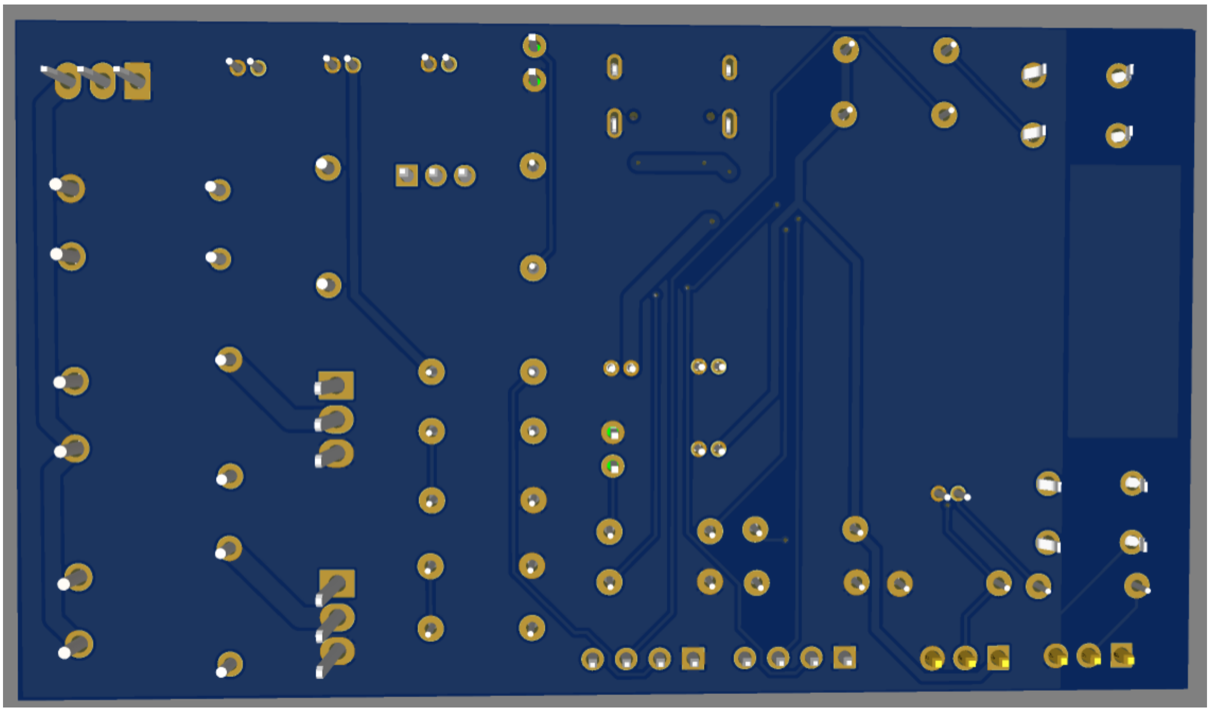

Figure 6: 3D Bottom View

6 Bill of Materials

The following table 2 lists the major components used in the Prosthetic Control Interface Board along with their quantities, footprints, and functional descriptions.

Table 2: Bill of Materials (BOM) of Prosthetic Control Interface Board

Component	Qty	Footprint	Description
10uF Capacitor	2	CAP-D4.0×F1.5	Power supply filtering capacitor
100nF Capacitor	5	CAP-D4.0×F1.5	Decoupling and noise filtering capacitor
1N5819 Schottky Diode	1	DO-41	Reverse polarity protection diode
1N4007 Diode	2	DO-41	Flyback protection diode for inductive loads
LED	2	LED0603-RD	Power and status indication
Fuse	1	1206	Input overcurrent protection fuse

Component	Qty	Footprint	Description
ESP32-WROOM-32	1	SMD Module	Main microcontroller and processing unit
CH340C	1	SOP-16	USB-to-UART communication interface
LD1117-3.3	1	SOT-223	3.3V voltage regulator
N-Channel MOSFET	2	TO-220 / SMD	Actuator and motor driver switching device
Push Button Switch	2	THT-6.0×6.0mm	Reset and boot control buttons
USB Type-C Connector	1	TYPE-C-31-M-12	USB programming and communication interface
Pin Headers	Multiple	2.54mm Header	UART, I2C, sensor, and output connectors
Resistors	Multiple	0603	Pull-up, pull-down, gate, and current limiting resistors

The complete BOM including manufacturer details, supplier part numbers, and pricing information was generated directly from EasyEDA and included separately in the project deliverables folder.

7 Design Decisions and Assumptions

This chapter discusses the important engineering decisions taken during the design process of the Prosthetic Control Interface Board. It also outlines the assumptions considered while developing the hardware architecture, PCB layout, and interfacing circuits.

7.1 Design Decisions

Several practical and engineering considerations were taken into account during the development of the PCB. The major design decisions are discussed below.

- ESP32 was selected due to its dual-core processing capability, integrated Wi-Fi/Bluetooth support, compact footprint, and rich GPIO availability.

- A dedicated ground plane was implemented on the bottom layer to reduce electrical noise and improve signal integrity.
- MOSFET-based actuator driving circuitry was used for reliable switching of external loads and motor control applications.
- USB Type-C connectivity was selected to provide modern, reversible, and reliable programming and communication support.
- Separate functional sections were maintained in the PCB layout to improve routing clarity and reduce analog signal interference.
- Wider traces were used for motor and power routing to safely handle higher current flow.

7.2 Assumptions

The following assumptions were considered during the hardware and PCB design process.

- The EMG and flex sensor modules provide conditioned analog output signals compatible with ESP32 ADC input levels.
- External motors and actuators operate within the safe current handling capability of the selected MOSFET devices and PCB traces.
- The USB interface is primarily intended for firmware uploading, serial debugging, and low-power communication.
- The PCB will operate under standard laboratory and prototype testing conditions.
- A regulated and stable DC power source is supplied to the board input.
- The ESP32 wireless antenna region remains free from copper pour and metallic obstruction to ensure proper RF performance.

8 Conclusion

The Prosthetic Control Interface Board was successfully designed as a compact and modular embedded hardware platform for prosthetic control applications. The PCB integrates sensor interfacing, embedded processing, actuator control, communication headers, and USB programming support into a single system.

The design incorporates EMG and flex sensor inputs for acquiring biological and motion-related signals, which are processed using the ESP32-WROOM-32 microcontroller. The processed signals can then be used to control external motors or actuators through MOSFET-based driver circuitry.

Special attention was given to PCB layout practices such as functional block placement, ground plane implementation, controlled routing, USB differential pair routing, and antenna keepout considerations. These measures improve signal integrity, reduce noise, and enhance overall system reliability.

The developed PCB serves as a prototype platform for future prosthetic and biomedical embedded system research. Further improvements such as advanced signal conditioning, wireless monitoring, battery management, and closed-loop control algorithms can be implemented in future versions of the design.

References

1. Espressif Systems, *ESP32-WROOM-32 Datasheet*, Version 3.6, 2023. Available: https://www.espressif.com

2. QinHeng Electronics (WCH), *CH340C USB to UART Converter Datasheet*, 2022. Available: http://www.wch.cn

3. STMicroelectronics, *LD1117 Low Dropout Voltage Regulator Datasheet*, 2021. Available: https://www.st.com

4. IPC Association, *IPC-2221: Generic Standard on Printed Board Design*, IPC Standards Documentation, 2012.

5. EasyEDA, *EasyEDA PCB Design Documentation*. Available: https://easyeda.com

6. USB Implementers Forum (USB-IF), *Universal Serial Bus Type-C Cable and Connector Specification*, Revision 2.2, 2022. Available: https://usb.org

7. Howard Johnson and Martin Graham, *High-Speed Digital Design: A Handbook of Black Magic*, Prentice Hall, 1993.

8. Douglas Brooks, *PCB Design Guidelines for Reduced EMI*, UltraCAD Design Inc., 2003.